2023-Winter Paper 2 – Urology

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Q1. Etiopathogenesis and management of Upper tract TCC

Type: LAQ (25 marks) (management also answered in 2016-Summer Paper 3, Q1)

TCC (transitional cell carcinoma) is taken as UTUC (upper tract urothelial carcinoma).

Introduction

- UTUC is 5–10% of urothelial cancers. Pelvicalyceal tumours are about twice as common as ureteric ones. Peak age 70–90; male:female 2:1.
- About 60% are invasive at diagnosis, because the upper tract has a thin muscle wall. Bladder recurrence follows in 22–47%.

Etiopathogenesis

Factor	Key facts and mechanism
Tobacco	Most important modifiable factor; risk 2.5–7-fold. Aromatic amines are activated to mutagenic N-hydroxylamines
Aristolochic acid	Balkan endemic nephropathy (UTUC 40% of renal tumours there) and Chinese herb nephropathy. Aristolactam–DNA adducts cause A:T → T:A transversions in TP53. Familial but **not inherited**; often bilateral
Lynch syndrome (HNPCC, hereditary nonpolyposis colorectal cancer)	Germline MMR (mismatch repair) mutation: MLH1, MSH2, MSH6, PMS2. Risk ~14-fold, highest with MSH2; found in 5–9% of screened UTUC
Occupation	"Amino tumours": benzidine, β-naphthylamine (dyes, rubber, petrochemicals); latency up to 20 years
Analgesic abuse	Phenacetin: papillary necrosis; capillarosclerosis is pathognomonic
Arsenic	Well water in south-west Taiwan (blackfoot disease)

- **Also know:** cyclophosphamide (acrolein); chronic infection and stones (squamous carcinoma); alcohol.
- **Lynch screening:** suspect if age <60, or a personal or family history of Lynch-spectrum cancer (colon, endometrium). Test the tumour for MSI (microsatellite instability) or MMR immunohistochemistry, then germline testing.
- **Pathogenesis:** two theories both operate. **Intraluminal seeding** (one clone spreading downstream) explains frequent bladder recurrence. **Field effect** (carcinogen in urine, independent clones) explains multifocality.
- Sequence: hyperplasia → dysplasia → CIS (carcinoma in situ). Variant histology in ~25%. Molecular subtypes include FGFR3-mutant (low grade), TP53/MDM2-mutant and

hypermutated (Lynch).

Diagnosis

- Haematuria (56–98%), flank pain (20–32%).
- **CTU (computed tomography urography):** best test (sensitivity ~92%). **Cystoscopy** always (concurrent bladder cancer 17%). Selective cytology.
- **Flexible URS (ureteroscopy) with biopsy:** grade is accurate in ~90%, but **stage cannot be assessed**. Staging is TNM (renal parenchymal invasion by a pelvic tumour = T3).

Risk stratification (EAU, European Association of Urology)

Low risk: all present	High risk: any one
Unifocal	Multifocal
Tumour <2 cm	Tumour >2 cm
Low-grade cytology and biopsy	High-grade cytology or biopsy
No invasion on CTU	Hydronephrosis
—	Previous radical cystectomy; variant histology

- AUA/SUO (American Urological Association/Society of Urologic Oncology) 2023 splits each group into favourable and unfavourable.

Management

A. Kidney-sparing surgery

- **Elective:** low-risk disease, whatever the other kidney. **Imperative:** solitary kidney, bilateral tumours, CKD (chronic kidney disease), unfit patient.
- **Ureteroscopic ablation:** biopsy, then holmium or thulium laser. Closed system; avoid circumferential ablation (stricture).
- **Percutaneous resection:** large pelvic or lower-pole tumours; risk of tract seeding.
- **Distal ureterectomy** with reimplantation (psoas hitch, Boari flap): gives a staging specimen.
- Adjuvant upper-tract BCG (bacillus Calmette–Guérin) or MMC (mitomycin C); **UGN-101** mitomycin gel for low grade.
- Lifelong endoscopic surveillance: upper-tract recurrence in about half.

B. RNU (radical nephroureterectomy) with bladder cuff: standard for high risk

- Clip the ureter early; kidney outside Gerota's fascia; never open the urinary tract; remove the **whole intramural ureter and cuff** (stump recurrence 30–75% if left); bag extraction.
- Open, laparoscopic and robotic routes are oncologically equivalent. Abandoned: endoscopic "pluck", laparoscopic stapling.
- **Template lymphadenectomy** for high risk: hilar plus paracaval (right) or para-aortic (left); pelvic nodes for lower ureter.
- **Single postoperative intravesical instillation:** ODMIT-C (MMC 40 mg) cut bladder recurrence 28% → 16% at 1 year.

C. Perioperative chemotherapy

- **POUT (2020):** adjuvant platinum after RNU for pT2–T4 or N+: 3-year DFS (disease-free survival) **71% vs 46%**. Gemcitabine–cisplatin; carboplatin if creatinine clearance 30–49.
- **Neoadjuvant** platinum when post-RNU renal function would rule out cisplatin.
- **Also know:** adjuvant nivolumab (CheckMate 274) if platinum-ineligible, but no clear benefit in the UTUC subgroup.

D. Metastatic disease (brief)

- First line: **enfortumab vedotin + pembrolizumab** (EV-302, 2023): overall survival ~31 vs ~16 months versus platinum chemotherapy. Alternative: gemcitabine–platinum, then avelumab maintenance.
- Later: pembrolizumab (KEYNOTE-045), erdafitinib for FGFR3 alterations. Node-positive: chemotherapy first, then consolidative surgery.

Follow-up

- After RNU: cystoscopy and cytology from 3 months; CTU yearly. After kidney-sparing: URS and CTU at closer intervals for life.

Diagram to draw: RNU specimen with bladder cuff, beside lymph node templates.

> **Key points to remember:**
> - Tobacco, aristolochic acid (A:T → T:A in TP53) and Lynch (MSH2) are the key causes.
> - EAU low risk needs every favourable feature; one adverse feature means high risk.
> - RNU with complete bladder cuff; single postoperative MMC (ODMIT-C 28% → 16%).
> - POUT: adjuvant platinum, 3-year DFS 71% vs 46%; EV-302 for metastatic disease.

Q2. Use of intestinal segments in urology

Type: LAQ (25 marks) (also answered as an SAQ in 2022-Summer Paper 1, Q3a)

Introduction

- Bowel is used to divert urine (conduit, continent pouch, neobladder), enlarge the bladder (augmentation), replace the ureter and make catheterisable channels.
- Choice depends on renal function, previous surgery or radiation, bowel disease, mesenteric length and the reconstruction needed.

Choice of segment

Segment	Electrolyte disturbance	Advantages	Drawbacks
Stomach	Hypochloraemic, hypokalaemic metabolic alkalosis	Secretes H^+ and Cl^-; less mucus; for renal failure, short gut, dense adhesions	**Haematuria–dysuria syndrome**, ulcers
Jejunum	Hyponatraemic, hypochloraemic, **hyperkalaemic acidosis**	Outside radiation field	Severe salt loss; avoid
Ileum	**Hyperchloraemic metabolic acidosis**	Mobile, constant blood supply; default conduit, ureteric replacement	B12 and bile-salt loss; obstruction ~10%
Colon	Hyperchloraemic metabolic acidosis	Easy antireflux tunnel; transverse colon outside pelvic radiation field	Diarrhoea if ileocaecal valve used; obstruction ~5%
Appendix	—	Mitrofanoff catheterisable channel	—

Technical principles

- **Preparation:** ERAS (enhanced recovery after surgery); routine mechanical bowel preparation not needed. Assess renal function.
- **Mesentery:** transilluminate, keep a major arcade, wide base; avoid watershed areas (Griffiths point at splenic flexure, Sudeck point at rectosigmoid).
- **Bowel anastomosis:** good blood supply, no tension, no spillage, close mesenteric window.
- **Detubularisation** (Kock): opening the bowel on its antimesenteric border and refolding it gives a spherical reservoir. This **abolishes coordinated contractions** and increases volume for the same surface area, so pressure stays low (Laplace's law).

Ureterointestinal anastomoses

Technique	Type	Stricture
Bricker	Refluxing, end to side	4–22% (mean 6%)
Wallace	Refluxing, ureters joined end to end	~3% (lowest)
Leadbetter–Clarke, Goodwin	Nonrefluxing tunnel in colon	8–14%

- Principles: spatulate, preserve periureteric tissue, tension-free, mucosa to mucosa, soft stents (leak 3–5% without). Bring the bowel to the ureter, not the ureter to the bowel.
- Fix and retroperitonealise the anastomosis. The left ureter passes behind the sigmoid mesentery; avoid angulation under the IMA (inferior mesenteric artery), a common stricture site.

- Reflux with normal unobstructed ureters has not been shown to harm the adult kidney; antireflux tunnels stricture more.

Applications

- **Incontinent diversion:** ileal conduit (10–15 cm ileum, 10–15 cm from ileocaecal valve); colon conduit after pelvic radiation.
- **Continent cutaneous pouch** (Indiana, Mainz, Kock) with appendix or tapered ileum channel.
- **Orthotopic neobladder** (Studer, Hautmann): about 50–70 cm detubularised ileum joined to the urethra.
- **Ureterosigmoidostomy:** now rare (acidosis, cancer).
- **Augmentation cystoplasty** ("clam" ileocystoplasty, ~25 cm ileum) for refractory neurogenic or small contracted bladder; needs CIC (clean intermittent catheterisation).
- **Ileal ureter:** isoperistaltic ileum for long defects. Contraindicated if creatinine >2 mg/dL, bladder outlet obstruction, inflammatory bowel disease or radiation enteritis.
- **Catheterisable channels:** Mitrofanoff appendicovesicostomy, Yang–Monti ileal tube.

Stomas

- **Protruding (nipple, "rosebud") stoma** for appliance wearers: less stenosis, better appliance fit. **Flush stoma** for continent pouches that are catheterised.
- **Loop-end (Turnbull) stoma** for obese patients with a short mesentery. Site is marked before surgery, through the rectus muscle.

Renal function needed for continent diversion

- Creatinine <2 mg/dL; GFR (glomerular filtration rate) >35 mL/min; urine pH ≤5.8 after ammonium chloride load; urine osmolality ≥600 mOsm/kg after water deprivation; minimal proteinuria.

Complications

Metabolic (worse with longer segment, longer dwell time, poorer renal function)

- **Hyperchloraemic acidosis (ileum, colon):** NH_4^+ replaces Na^+ on the Na^+/H^+ exchanger, coupled with Cl^-/HCO_3^- exchange, so ammonium chloride is absorbed. Treat with sodium bicarbonate or citrate; chlorpromazine or nicotinic acid limit chloride transport.
- **Hypokalaemia:** worse with colon; replace potassium before correcting acidosis (paralysis risk).
- **Jejunal syndrome:** treat with sodium chloride and thiazide. **Gastric alkalosis:** H2 blockers or PPI (proton-pump inhibitor).
- **Altered sensorium:** hyperammonaemia (drain the pouch, lactulose), hypomagnesaemia.
- **Drug reabsorption:** phenytoin, methotrexate; drain pouches during chemotherapy.
- **Osteomalacia:** acidosis, vitamin D resistance, calcium loss; growth retardation in children.
- **Nutrition:** B12 deficiency (give parenteral B12 after 5–7 years of ileal diversion), bile-salt diarrhoea.

Other

- Infection (bacteriuria in about three-quarters of conduits; treat pure Proteus or Pseudomonas).
- **Stones:** colon conduit 3–4%, ileal conduit 10–12%, continent reservoir ~20% (mucus, staples).
- **Augment perforation:** surgical emergency; regular CIC prevents it.
- Ureterointestinal stricture (lifelong risk), renal deterioration, bowel obstruction, stomal stenosis and parastomal hernia.

- **Cancer:** ureterosigmoidostomy 6–29% after 10–20 years (endoscopy from 5 years); augments (endoscopy from 10 years).

Follow-up

- Lifelong: electrolytes, bicarbonate, renal function, B12, upper-tract imaging (ultrasound or CT) for stricture and stones; endoscopy of colonic or augmented segments as above.

Recent advances

- Robotic intracorporeal diversion (complications similar to open in experienced hands); tissue-engineered bladder (experimental).

Diagram to draw: ileal conduit with mesenteric arcade and Wallace anastomosis.

> **Key points to remember:**
>
> - Ileum and colon: hyperchloraemic acidosis (NH_4Cl absorption).
> - Jejunum: hyperkalaemia; stomach: hypochloraemic alkalosis.
> - Continent diversion: GFR >35, pH ≤5.8, osmolality ≥600.
> - Wallace has the lowest stricture rate; detubularisation gives low pressure.

Q3a. GFR clinical assessment

Type: SAQ (10 marks) (donor setting answered in 2019-Summer Paper 1, Q3a)

Definition

- GFR (glomerular filtration rate) is the plasma volume filtered by both kidneys per minute; normal about 120 mL/min/1.73 m^2. It is the standard index of kidney function. The ideal marker is freely filtered and neither secreted nor reabsorbed.

Methods

1. Serum creatinine and eGFR (estimated GFR)

- Creatinine comes from muscle; freely filtered but also secreted, so it **overestimates GFR** as function falls.
- Varies with **muscle mass, diet and age**; trimethoprim blocks secretion.
- **CKD-EPI (Chronic Kidney Disease Epidemiology Collaboration) 2021 equation:** age and sex, **no race term**; currently recommended. MDRD (Modification of Diet in Renal Disease) and Cockcroft–Gault are older.

2. Cystatin C

- 13-kDa protein from all nucleated cells; freely filtered, fully reabsorbed and broken down. Independent of muscle mass; affected by thyroid state, steroids, inflammation.
- **Combined creatinine–cystatin C equation** is the most accurate estimate.

3. 24-hour creatinine clearance

- (Urine creatinine × urine volume) ÷ serum creatinine. Collection errors; secretion causes overestimation.

4. mGFR (measured GFR): reference

Marker	Method
Inulin	Urinary clearance; gold standard, research only
Iohexol, iothalamate	Plasma or urinary clearance
^{51}Cr-EDTA (ethylenediaminetetraacetic acid)	Plasma clearance
^{99m}Tc-DTPA (diethylenetriaminepentaacetic acid)	Plasma sampling or gamma camera (Gates method); also gives **split function**

Clinical use

- **CKD (chronic kidney disease) staging (KDIGO, Kidney Disease: Improving Global Outcomes):** G1 ≥90, G2 60–89, G3a 45–59, G3b 30–44, G4 15–29, G5 <15 mL/min/1.73 m^2, always with albuminuria (ACR, albumin-to-creatinine ratio).
- **Living donor (KDIGO 2017):** ≥90 accept, 60–89 individualise, <60 decline; confirm with mGFR.
- **Urology:** drug and contrast dosing, cisplatin eligibility, continent diversion (>35 mL/min), partial versus radical nephrectomy.
- **Acute change:** eGFR is invalid when function is unstable; AKI (acute kidney injury) uses creatinine change and urine output. NGAL (neutrophil gelatinase-associated lipocalin) is emerging.

Diagram to draw: plasma tracer clearance curve after a single injection.

> **Key points to remember:**
>
> - Creatinine overestimates GFR (secretion); CKD-EPI 2021 is race-free.
> - Creatinine–cystatin C combined equation is most accurate.
> - Inulin is gold standard; DTPA gives GFR and split function.

Q3b. Neuromodulation

Type: SAQ (10 marks) (also answered in 2019-Summer Paper 4, Q1)

Taken as neuromodulation for lower urinary tract dysfunction.

Definition and mechanism

- Electrical stimulation of a peripheral nerve that changes central reflex pathways controlling the bladder. It acts on **afferents**, not directly on muscle.
- **OAB (overactive bladder):** afferent input damps PMC (pontine micturition centre) firing, so fewer involuntary contractions.
- **Non-obstructive retention:** restores awareness of filling and relaxes the external sphincter. Hence one therapy treats opposite disorders.
- Exclude anatomical causes first (obstruction, stones, cancer).

Modalities

Modality	Method	Main use
SNM (sacral neuromodulation)	Tined lead in **S3** foramen + implanted generator	Refractory OAB, NOUR (non-obstructive urinary retention), faecal incontinence
PTNS (percutaneous tibial nerve stimulation)	34-gauge needle above medial malleolus	Refractory OAB
Implantable tibial devices	Small device over tibial nerve	OAB without clinic visits
Pudendal stimulation	Lead at Alcock's canal	SNM failure (off-label)
SARS (sacral anterior root stimulator)	Anterior S2–S4 roots + dorsal rhizotomy	Complete spinal cord injury

Sacral neuromodulation

- **Third-line** for OAB (AUA, American Urological Association; EAU, European Association of Urology).
- **Test phase:** PNE (percutaneous nerve evaluation, 5–7 days) or staged tined lead (up to 4 weeks). **Implant if ≥50% improvement** in diary symptoms.
- **S3 response:** perineal "bellows" plus great-toe flexion.
- **Evidence:** InSite: success 61% vs 42% with medical therapy. ROSETTA: onabotulinumtoxinA 200 U slightly better for UUI (urgency urinary incontinence) but more UTI (urinary tract infection) and self-catheterisation.
- **Problems:** pain, lead migration, loss of effect; revision >30% at 5 years; cost. Newer rechargeable, MRI-conditional devices.

Tibial nerve stimulation

- Tibial nerve carries **S2–S3** fibres. Needle 3–4 cm above medial malleolus; toe flexion confirms placement; **30 min weekly × 12**, then maintenance.

- **SUmiT:** response 55% vs 21% with sham.
- **Also know:** IVES (intravesical electrical stimulation) in children; dorsal genital nerve stimulation.

Diagram to draw: S3 lead with buttock generator, and PTNS needle above the medial malleolus.

> **Key points to remember:**
> - Afferent modulation explains benefit in both OAB and retention.
> - S3: bellows + great-toe flexion; implant if ≥50% improvement.
> - PTNS: 12 weekly sessions; SUmiT 55% vs 21%.

Q3c. Post transplant malignancy

Type: SAQ (10 marks)

Introduction

- Cancer risk after kidney transplantation is about 2–3 times the general population; a leading cause of late death with a functioning graft.

Causes

- **Lost immune surveillance** from immunosuppression (dose and duration matter; depleting antibodies add risk).
- **Oncogenic viruses:** EBV (Epstein–Barr virus) → PTLD (post-transplant lymphoproliferative disorder); HHV-8 (human herpesvirus-8) → Kaposi sarcoma; HPV (human papillomavirus) → skin and anogenital cancer; hepatitis B/C → liver cancer.
- **Direct drug effects:** azathioprine sensitises skin to UV (ultraviolet) light; calcineurin inhibitors promote tumour growth. mTOR (mammalian target of rapamycin) inhibitors are **antitumour**.
- **Pre-existing risk:** ESRD (end-stage renal disease), acquired cystic disease, analgesic nephropathy; rarely donor-transmitted.

Common cancers

Cancer	Key facts
Non-melanoma skin	Commonest; squamous exceeds basal cell (reverse of general population); multiple, aggressive
PTLD	Mostly B-cell, EBV-driven; highest in first year, children and **EBV-negative recipient of EBV-positive donor**; may present as graft mass
Kaposi sarcoma	HHV-8; regresses with less immunosuppression and switch to sirolimus
RCC (renal cell carcinoma)	4.6% of post-transplant cancers; **90% in native kidneys**; graft tumours mostly papillary, low grade
Bladder cancer	About **3-fold** risk; more aggressive
Prostate	Same incidence as general population

Management

- **Reduce immunosuppression** and switch to an mTOR inhibitor; sirolimus conversion reduced new skin squamous cancers (Euvrard, 2012).
- **PTLD:** reduce immunosuppression, rituximab, then chemotherapy (R-CHOP).
- RCC, prostate and bladder: treat as in the general population; protect graft vessels and ureter at surgery. BCG (bacillus Calmette–Guérin) is usable in selected bladder cancer.

Prevention and screening

- Pre-transplant cancer screening and waiting periods (none after curative treatment of small RCC <3 cm, Gleason 6 prostate cancer, superficial bladder cancer).
- Sun protection, yearly skin examination, age-appropriate screening.
- KDIGO (Kidney Disease: Improving Global Outcomes) does **not** endorse routine native-kidney screening; screen only high-risk recipients.

Diagram to draw: pie chart of post-transplant cancers (skin largest).

Key points to remember:

- Skin cancer commonest; squamous more than basal cell.
- PTLD: EBV-negative recipient of EBV-positive donor, first year.
- RCC mostly in native kidneys; bladder cancer 3-fold and aggressive.
- Reduce immunosuppression and switch to mTOR inhibitor.

Q3d. Biomarkers for patients with advanced RCC

Type: SAQ (10 marks)

Introduction

- A **prognostic** biomarker predicts outcome regardless of treatment; a **predictive** biomarker predicts benefit from a specific drug.
- In advanced RCC (renal cell carcinoma), only clinical risk models are used routinely. No molecular biomarker is yet validated to select treatment (EAU, European Association of Urology; NCCN, National Comprehensive Cancer Network).

Clinical and laboratory: risk models

Factor	MSKCC (Memorial Sloan-Kettering Cancer Center)	IMDC (International Metastatic RCC Database Consortium)
Karnofsky performance status <80%	Yes	Yes
Haemoglobin < lower limit of normal	Yes	Yes
Corrected calcium high	>10 mg/dL	> upper limit of normal
LDH (lactate dehydrogenase) >1.5 × normal	Yes	No
Time factor	No prior nephrectomy	<1 year from diagnosis to treatment
Neutrophils, platelets > normal	No	Yes

- IMDC: favourable 0 factors (median survival 43 months), intermediate 1–2 (22.5 months), poor ≥3 (7.8 months).
- IMDC group guides first-line choice (nivolumab–ipilimumab not for favourable risk).
- **Also know:** neutrophil-to-lymphocyte ratio, CRP (C-reactive protein), liver or bone metastases (poor prognosis).

Tissue biomarkers

- **PD-L1 (programmed death-ligand 1):** benefit of checkpoint combinations is **independent of PD-L1**, so it is not used to select patients.
- **Sarcomatoid features:** respond well to checkpoint inhibitors.
- **Gene expression signatures:** IMmotion150 showed high T-effector signature tumours do better with atezolizumab–bevacizumab; high angiogenic signature tumours do well with sunitinib. Investigational.
- **Genomic:** BAP1 loss and CDKN2A loss indicate poor prognosis; PBRM1 data conflicting. MET alterations in papillary RCC (cabozantinib); FH (fumarate hydratase) deficiency. TMB (tumour mutational burden) is low and not predictive.

Circulating and other biomarkers

- **KIM-1 (kidney injury molecule-1):** high serum levels predict recurrence and benefit from adjuvant immunotherapy (investigational).
- **ctDNA (circulating tumour DNA):** RCC sheds little DNA; methylation assays are promising.
- **Also know:** antibiotics during checkpoint therapy reduced survival (microbiome); sunitinib-induced hypertension is linked to better outcome.

Diagram to draw: IMDC risk groups with survival curves.

Key points to remember:

- IMDC: 6 factors; poor risk ≥3; LDH only in MSKCC.
- PD-L1 does not select patients in RCC.
- Sarcomatoid tumours respond well to immunotherapy.

Q3e. Pre transplant Immunosupression treatment options

Type: SAQ (10 marks) (maintenance drugs answered in 2020-Summer Paper 4, Q1)

Taken as induction immunosuppression and desensitisation given before or at transplantation.

Principle

- Rejection risk is highest in the first weeks. **Induction** gives intense immunosuppression around transplant; **desensitisation** removes preformed antibody in HLA (human leucocyte antigen)- or ABO-incompatible recipients.
- Immunological risk is assessed first: cPRA (calculated panel-reactive antibody), DSA (donor-specific antibody) on Luminex beads, crossmatch (CDC, complement-dependent cytotoxicity, and flow).

Induction agents

Agent	Mechanism	Use and drawbacks
Basiliximab	Anti-CD25 (IL-2, interleukin-2, receptor); non-depleting	Low-risk recipients; 20 mg on days 0 and 4; few side effects
rATG (rabbit antithymocyte globulin)	Polyclonal T-cell depletion	High-risk (sensitised, retransplant, delayed graft function); less rejection than basiliximab; cytokine release, leucopenia, infection, PTLD (post-transplant lymphoproliferative disorder)
Alemtuzumab	Anti-CD52; profound depletion	Allows steroid avoidance; off-label
IV methylprednisolone	Cytokine suppression	Given intraoperatively with all regimens

- Over 90% receive induction, then **tacrolimus + mycophenolate** ± steroids. In living-donor transplants, tacrolimus and mycophenolate often start days before surgery.

Desensitisation: HLA-incompatible

- **Plasmapheresis** (removes antibody) plus **low-dose IVIG (intravenous immunoglobulin)** after each session; or **high-dose IVIG** (immunomodulation).
- **Rituximab** (anti-CD20) reduces B cells that renew antibody. Bortezomib (plasma cells) and eculizumab (anti-C5) in selected cases.
- **Imlifidase:** bacterial enzyme that **cleaves all IgG** within hours, turning a positive crossmatch negative; EMA (European Medicines Agency) conditional approval 2020 for highly sensitised patients.
- Desensitised live-donor transplant gives better survival than staying on dialysis (Montgomery, 2011).

Desensitisation: ABO-incompatible

- **Rituximab** 2–4 weeks before; **plasmapheresis or immunoadsorption** to lower anti-A/B titres to a centre target (commonly ≤1:8); IVIG; tacrolimus–mycophenolate started before surgery. Splenectomy is now obsolete.
- Post-transplant titre monitoring; A2 donors carry lower risk.

Alternatives

- **Kidney paired exchange** avoids desensitisation; acceptable-mismatch programmes.

Diagram to draw: antibody titre falling with plasmapheresis sessions to target before transplant.

Key points to remember:

- Basiliximab for low risk; rATG for high immunological risk.
- HLA-incompatible: plasmapheresis + IVIG ± rituximab; imlifidase cleaves IgG.
- ABO-incompatible: rituximab + plasmapheresis to target titre.

Q3f. Physiology of UPJ in relation to urine transport

Type: SAQ (10 marks)

UPJ = ureteropelvic (pelviureteric) junction.

Anatomy relevant to function

- The UPJ is the apex of a funnel-shaped pelvis; its spiral smooth muscle is continuous with calyces and ureter.

Pacemaker and electrical activity

- Peristalsis is **myogenic**: the excised or denervated ureter keeps contracting. Nerves carry pain and only modulate.
- **Primary pacemaker: atypical smooth muscle cells** at the pelvicalyceal border; they fire spontaneously (HCN, hyperpolarisation-activated cyclic nucleotide-gated, channels).
- **ICC-like (interstitial cells of Cajal-like) cells** (c-Kit positive) in the pelvis and UPJ relay the signal to typical smooth muscle and act as back-up pacemakers.
- Signal spreads through **gap junctions**. Calcium entry gives a plateau action potential; calcium–calmodulin activates myosin light-chain kinase, causing contraction.

Bolus formation and transport

1. Urine fills the pelvis at low pressure; the UPJ is closed.
2. Pelvic contraction pushes urine through the opening UPJ into the upper ureter.
3. The UPJ closes behind it, forming a **bolus**.
4. Ureteric walls **coapt** behind the bolus; the wave (2–6 per minute, 20–80 cm H_2O) carries it to the bladder.

- At **low flow**, the UPJ acts as a filter: several pelvic contractions occur for each ureteric contraction.
- As **flow rises**, ureteric frequency reaches 1:1, then bolus volume rises.
- At **very high flow**, boluses merge into a continuous column and the ureter dilates **without obstruction**.

Effect of obstruction

- **Acute:** stronger, more frequent contractions.
- **Chronic:** dilatation, walls fail to coapt, peristalsis becomes ineffective; raised pelvic pressure lowers GFR (glomerular filtration rate).
- **Intrinsic UPJO (UPJ obstruction):** abnormal muscle, excess collagen, fewer ICC-like cells, so an **aperistaltic segment** cannot transmit the wave.
- **Also know:** endotoxin and progesterone reduce peristalsis. Diuretic renogram and Whitaker test (chapter: pressure difference >12 cm H_2O suggests obstruction; classic teaching: <15 unobstructed, >22 obstructed) assess transport.

Diagram to draw: pelvis, UPJ and ureter showing bolus formation and coaptation.

Key points to remember:

- Myogenic peristalsis from atypical smooth muscle pacemakers; ICC-like cells back up.
- Low flow: several pelvic beats per ureteric beat; very high flow: dilatation without obstruction.
- Intrinsic UPJO: aperistaltic segment with fewer ICC-like cells.

Rapid Revision Sheet – 2023-Winter Paper 2

All the key points from this paper in one place. Read this in the last 30 minutes before the exam.

Q1. Etiopathogenesis and management of Upper tract TCC (LAQ)

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- EAU low risk needs every favourable feature; one adverse feature means high risk.
- RNU with complete bladder cuff; single postoperative MMC (ODMIT-C 28% → 16%).
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Q2. Use of intestinal segments in urology (LAQ)

- Ileum and colon: hyperchloraemic acidosis (NH_4Cl absorption).
- Jejunum: hyperkalaemia; stomach: hypochloraemic alkalosis.
- Continent diversion: GFR >35, pH ≤5.8, osmolality ≥600.
- Wallace has the lowest stricture rate; detubularisation gives low pressure.

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- Creatinine overestimates GFR (secretion); CKD-EPI 2021 is race-free.
- Creatinine–cystatin C combined equation is most accurate.
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- Afferent modulation explains benefit in both OAB and retention.
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- Skin cancer commonest; squamous more than basal cell.
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- RCC mostly in native kidneys; bladder cancer 3-fold and aggressive.
- Reduce immunosuppression and switch to mTOR inhibitor.

Q3d. Biomarkers for patients with advanced RCC (SAQ)

- IMDC: 6 factors; poor risk ≥3; LDH only in MSKCC.
- PD-L1 does not select patients in RCC.
- Sarcomatoid tumours respond well to immunotherapy.

Q3e. Pre transplant Immunosupression treatment options (SAQ)

- Basiliximab for low risk; rATG for high immunological risk.
- HLA-incompatible: plasmapheresis + IVIG ± rituximab; imlifidase cleaves IgG.
- ABO-incompatible: rituximab + plasmapheresis to target titre.

Q3f. Physiology of UPJ in relation to urine transport (SAQ)

- Myogenic peristalsis from atypical smooth muscle pacemakers; ICC-like cells back up.
- Low flow: several pelvic beats per ureteric beat; very high flow: dilatation without obstruction.
- Intrinsic UPJO: aperistaltic segment with fewer ICC-like cells.